Measuring Progress on Long-Tailed Dental Radiograph Segmentation: Leakage, Ceilings, and Noise Floors

Abstract

We evaluate detection and instance segmentation on a 31-class panoramic dental radiograph corpus across more than twenty-five controlled configurations. Every change to the objective, the sampling, the loss weighting or the mask head lands within seed-level variation. The one change that clears every check is physical rather than algorithmic: restoring the input resolution the corpus natively carries is worth $+14.2\,\%$ relative segmentation mAP, with the gain concentrated in the small clinical findings the objective-level arms were built to help. Locating that lever, and certifying it, required the apparatus this paper describes.

The contribution is not the null itself but the apparatus required to establish it, each element of which caught a confident wrong answer that would otherwise have been reported. **(1)** The corpus cannot be evaluated as distributed: 95.9 % of its test partition shares source images with training. **(2)** Perceptual hashing alone is not a leakage test on this modality, with a false-positive rate near 100 %. **(3)** The mask representation imposes a measurable ceiling that is *not* the binding constraint; a higher-resolution head with 878,432 *fewer* parameters performs 0.72 pp worse, as predicted. **(4)** Distance metrics conditioned on per-model subsets invert their own sign once compared on the intersection of scored cases. **(5)** Group means over low-support classes fabricate results: in one case 98 % of a $+0.53$ pp mAP gain came from a single class with *two* test instances. **(6)** Segmentation AP dissociates from mask quality in both directions: one arm gains $+0.64$ pp entirely from the detection side, and a query-based architecture gains $+2.16$ pp while every paired mask metric on 5,460 common cases is separably *worse*. **(7)** Cross-dataset AP is meaningless when two corpora annotate a disease at different granularity: our caries detector scores exactly 0.0000 on an external benchmark that outlines the affected tooth where we outline the lesion, a $19\times$ area ratio, while the same predictions are 74 % precise once the question is asked at tooth level. **(8)** The published logit-adjustment constant does not transfer from $\sim$100:1 to 34,320:1 imbalance.

A closed-form oracle fit attributes 85 % of the mask deficit to coefficient prediction rather than to the learned basis. Acting on that attribution fails in both available directions—capacity leaves masks unchanged, and direct coefficient supervision makes them worse—so the deficit is correctly localised and not thereby actionable. Finally, a fifteen-model consensus bounds the corpus itself: 911 pathology annotations, 26–43 % of the floor classes, are invisible to every model trained here, and the model that lifted caries by 35 % recovers 2.7 % of them. The remaining ceiling is the labels, and the audit list is generated rather than hypothesised.

1 Introduction

Long-tailed instance segmentation and detection on clinical imagery is an active area, and the standard toolkit—class-balanced losses, logit adjustment, repeat-factor sampling, boundary-aware objectives—is well established on natural-image benchmarks. We set out to apply that toolkit to a 31-class dental radiograph corpus with 34,320:1 imbalance and measure the contribution of each component.

None of it worked. That is worth reporting, because the reasons are specific, measurable, and mostly not about the methods. The corpus as distributed cannot support evaluation at all. The architecture imposes a ceiling that is real but not binding. The metrics the field uses for contour quality invert their sign under a denominator that differs per model. And group means over classes with single-digit support will manufacture a headline on demand—twice, in our case, before we caught them.

We regard contributions (2), (4) and (5) as the most transferable: each is a measurement protocol that any group working on long-tailed clinical segmentation can apply, and each caught an error in our own work that would have survived review.

2 Related Work

Long-tailed recognition. Class-balanced loss weighting via effective number of samples [2] and logit adjustment [11] are the standard remedies. Kang et al. [6] showed representation and classifier learning should be *decoupled*, with rebalancing applied at the classifier stage; Wang et al. [15] adapt this to instance segmentation. Repeat-factor sampling [4] rebalances at the data level, and Seesaw loss [14] rescales gradients per class pair.

Our results bear on the *regime* these were validated in. Menon et al. [11] tune $\tau = 1.0$ on CIFAR-LT and ImageNet-LT at roughly 100:1 to 1000:1 imbalance. At 34,320:1 the same constant applies a $+11.47$ logit shift and destroys the classifier. We find the failure monotonic in τ, which localises the problem to magnitude rather than principle.

High-quality mask prediction. PointRend [10] recasts segmentation as rendering, refining adaptively at uncertain points. RefineMask [16] fuses fine-grained features in stages. Mask Transfiner [8] represents masks as a quadtree and corrects only error-prone nodes. All three exist because coarse mask prediction is a known failure mode. We measure that failure mode directly rather than assuming it, and find it is *not* what limits our model—a result invisible without the model-free ceiling measurement of Section 6.

image files	13,932
distinct source images	6,395
test files whose source appears in train/valid	**1,516 / 1,580 (95.9 %)**
genuinely unseen test sources	**64**
patients in both train and test	194

Table 1: The distributed split. Augmentation was applied before partitioning rather than after, so any metric computed on it principally measures memorisation.

check	impl. A	impl. B
exact duplicate groups (SHA-256)	0	0
NCC-confirmed near-duplicates	**0**	**0**
dHash candidates examined	441,240	792,811
rejected as look-alikes	all	all
scope	test vs rest	all pairs

Table 2: Disjointness verified twice by independent implementations. Maximum NCC across 792,811 candidate pairs is 0.9334, against a 0.98 confirmation threshold.

Boundary-aware objectives. Boundary loss [9] integrates the prediction against a precomputed distance map; HD loss [7] estimates Hausdorff distance directly; Boundary IoU [1] is an evaluation metric restricted to a contour band. Our band-Dice objective is a synthesis of known components—morphological gradient as a differentiable edge extractor, soft Dice [12] as the overlap objective—and we say so rather than claiming a new loss family.

Augmentation. Copy-paste [3] reliably helps on natural images. We find it harmful here, and propose a mechanism: in panoramic radiographs anatomical position is part of the class definition, so a finding pasted into an anatomically impossible location teaches the model to detect the paste rather than the pathology.

3 The Corpus Cannot Be Evaluated as Distributed

Two further defects: a spurious `croen` category at id 12 in the test COCO file shifts every higher class id out of alignment with train and valid, and `data.yaml` points `test:` at the validation images.

Replacement. Patient-grouped and class-stratified: 9752 / 2090 / 2090, with test coverage rising from 13 of 31 classes to 29.

3.1 Perceptual Hashing Alone Is Not a Leakage Test

On natural images a dHash Hamming distance below $\sim$5 bits is strong evidence of duplication. On panoramic radiographs it is nearly meaningless: every image is the same anatomy, same framing, same grayscale statistics, so unrelated studies routinely fall within 5 bits.

Run without pixel confirmation, dHash reports **792,811 leaking pairs on a split we independently verify as clean**—a false-positive rate near 100 %. Any audit of this modality that stops at perceptual hashing will condemn a clean split, or worse, be tuned until it stops complaining.

Two-stage detection—dHash as a cheap candidate generator, normalised cross-correlation as confirmation—separates the cases cleanly. Genuine re-encoded duplicates score $\mathrm{NCC} \geq 0.99$; distinct studies stay below 0.94.

prototype grid	mean Dice	cannot reach IoU 0.75
160×160 **(stock)**	**0.8963**	**17.7 %**
320×320	0.9492	5.4 %
640×640	0.9901	0.5 %

Table 3: Model-free ceiling over 20,601 instances. Root canal treatment is capped at IoU 0.622 and caries at 0.721—both *below* the 0.75 threshold, excluding 4,493 instances from AP75 before training begins.

3.2 Statistical Power

Sixteen of 31 classes appear in fewer than 10 validation images; two appear in none; eleven tail classes score exactly zero. **Image counts, not instance counts, are the effective sample size.** No long-tail claim is supportable on this corpus, and we make none.

4 The Representation Ceiling, and Why It Is Not the Constraint

YOLOv8-seg predicts 32 coefficients per instance and reconstructs masks as a linear combination of prototypes at input$/4$—160×160 at $\mathrm{imgsz} = 640$. The median instance here is **6 px on that grid**; 68 % are under 8 px.

We bound what any model could achieve by round-tripping every ground-truth mask through the grid and scoring it against itself.

The model's behaviour agrees: mask AP50 reaches 87 % of box AP50, mask AP75 only 49 %.

4.1 The Inference We Initially Drew Was Wrong

This looks like a diagnosis. It is not.

ceiling at the current grid	0.8963
achieved by the model	**0.6969**
headroom already available	0.1994
additional headroom from doubling	0.0529

Table 4: The model sits at 78 % of the ceiling it already has.

metric	ref.	bnd.	difference	sep.
Dice	0.6953	0.6969	$+0.0017$	no
IoU	0.5685	0.5706	$+0.0021$	no
boundary F	0.7857	0.7881	$+0.0024$	no
HD95	66.36	68.66	$\mathbf{+2.30}$ **(worse)**	**yes**
ASSD	18.40	18.82	$+0.43$ (worse)	no

Table 5: Recomputed on the 5,385 cases both models scored, paired, with the per-case difference bootstrapped over images. Correcting the denominator did not attenuate the effect—it *inverted the sign*.

We tested it. A prototype head fed from P2 (stride 4) rather than P3 (stride 8) yields 320×320 prototypes from genuine high-resolution features, using sub-pixel convolution [13] to keep the wide tensor off the high-resolution grid, with **878,432 fewer parameters** than stock so any gain could not be a capacity artifact. Against a matched control at identical budget and protocol: -0.72 **pp mAP,** -0.49 **pp AP75**.

The ceiling is real and worth reporting. It is not the binding constraint, and the check that establishes this—comparing achieved performance against the ceiling—costs one line of arithmetic. We ran it after the experiment. It should be run before.

5 Four Metric Failures That Produce Confident Wrong Answers

5.1 Distance Metrics on Per-Model Subsets

HD95 and ASSD are undefined when either mask is empty, so implementations average them over cases where both are non-empty. **That denominator differs per model.** A model that misses a hard structure drops that case from its own average and can post better distances by predicting less.

Averaged per model, our boundary objective appeared **17.3 % better on HD95 and 24.6 % better on ASSD, with disjoint confidence intervals**—the strongest result in the project.

Protocol. Any metric conditioned on a per-model subset must be compared on the intersection of cases both models scored, paired, with coverage reported alongside. Reporting distances without coverage permits a model to win by predicting less.

5.2 Group Means over Low-Support Classes

Two headline gains, in two independent projects, dissolved under decomposition. For detection, decoupled classifier retraining gave $+0.53$ pp mAP and $+1.25$ pp tail—positive on every metric, with the largest gain exactly where a long-tail method should deliver.

Protocol. Report per-class deltas with support counts alongside any group mean, and state the result restricted to classes with adequate support. A group AP over classes with single-digit support is not a measurement.

observed mAP delta	$+0.53$ pp
largest contributor	one class, **2 test instances**
its contribution	$+0.52$ pp
share of total	**98 %**
excluding that class	$+0.01$ **pp**
on 18 classes with $\geq$10 instances	6 improve, 12 worsen

Table 6: Decomposition of the detection gain.

5.3 Segmentation AP Is a Conjunction, and Rises on Detection Alone

The largest single gain in this study is $+0.64$ pp of segmentation mAP, from widening the mask-coefficient branch `cv4` from 80 to 256 hidden channels ($0.29 \rightarrow 3.86$ M parameters in the branch, nothing else changed). Against a measured noise floor of ± 0.21 pp it clears the threshold threefold, and tail AP nearly triples, from 0.0101 to 0.0297. Reported as it stands it is the one positive architectural result in the work.

It is not a mask result. Scoring the *same* predictions as boxes instead of masks gives $+0.74$ pp, *larger* than the segmentation gain.

arm	segm mAP	Δ	box mAP	Δ
reference	0.1055	—	0.1551	—
cv4 widened	**0.1118**	$\mathbf{+0.64}$	0.1625	$\mathbf{+0.74}$

Table 7: The same predictions, scored two ways. `cv4` emits mask coefficients and has no path to the box head, so the gain cannot originate there.

Segmentation AP is a conjunction: an instance counts only if the detection matches *and* the mask clears the IoU threshold. A change that improves only the detector therefore raises it while leaving masks untouched. Paired on the 5352 cases where both models emit a mask, Dice moves -0.0016 with the interval spanning zero, and the one metric separable from zero is boundary F at -0.0043—in favour of the *reference*. Rebuilding the branch with three times the parameters perturbs the gradients returning into the shared neck; the detector improves as a side effect, and the masks it was meant to fix do not.

A second, stronger instance arrived later and in the opposite direction. A query-based detector (Mask DINO, R50) posts the highest segmentation mAP in the study, 0.1271 against the baseline's 0.1055, while *every* paired mask-quality metric on the 5460 common cases is separably worse: Dice -0.039, IoU -0.035, boundary F -0.065, HD95 $+20.7$ px, ASSD $+5.3$ px, all with intervals excluding zero and all favouring the baseline. Its box mAP is also no better (0.1503 against 0.1551). The headline metric is lifted by ranking and recall behaviour across the score distribution, not by the quality of any mask it draws. Segmentation AP and mask quality therefore dissociate in both directions: an arm can gain AP from detection alone (above), and an architecture can gain AP while drawing measurably worse masks.

Protocol. Any claim that an architectural change improved segmentation must report the same predictions scored as boxes. If the box delta is comparable to or larger than the mask delta, the change is not a mask result whatever the segmentation AP says. This costs one extra evaluation pass and is the only cheap way to separate the two factors the metric multiplies together.

5.4 Cross-Dataset AP When Annotation Granularity Differs

The three failures above are internal to one corpus. The fourth appears the moment a second corpus is introduced, and it is the most emphatic of the four because the wrong answer it produces is exactly zero.

We converted the DENTEX 2023 disease subset [5] into our vocabulary: 678 panoramic radiographs, 3529 professionally annotated instances, of which three classes map onto ours. Scoring our baseline on it zero-shot gives:

class	DENTEX box AP	DENTEX mask AP
impacted tooth	0.2902	0.2954
Caries	**0.0000**	**0.0000**
Periapical lesion	**0.0000**	**0.0000**

Table 8: Zero-shot, and two thirds of it is an artefact.

Read literally, the detector does not work at all on external data. It is not a defect of the pipeline: the model emits 26,340 caries detections on these images, the category mapping is verified, and impacted tooth reaches an IoU of 0.911 on the same files, so loading, coordinates and class identity are all sound. Yet across 4367 caries detections and 2767 caries ground truths, no pair anywhere exceeds **IoU 0.384**.

The cause is that the two corpora annotate the same disease at different granularity.

class	DENTEX median box	ours	ratio
Caries	1.228 % of image	0.064 %	19.2×
Periapical lesion	1.407 %	0.150 %	9.4×
impacted tooth	1.072 %	0.895 %	1.2×

Table 9: Within DENTEX a caries box is 1.15× the area of its own impacted-tooth box, so it marks the affected *tooth*; ours is 0.07×, so it marks the *lesion*.

DENTEX follows a quadrant–enumeration–diagnosis pipeline, so a diagnosis is a property of a tooth and the box is the tooth. Our corpus outlines the lesion. A box one nineteenth of the area of its target cannot reach IoU 0.5 however perfectly it is placed, so the zero is arithmetic rather than empirical. Impacted tooth transfers only because it is the one class on which the two corpora agree about what is being outlined, which is precisely why it was chosen as the control.

Asking the granularity-correct question instead—does our lesion detection fall inside the affected-tooth box—the same predictions give:

class	conf	recall	precision
Caries	0.15	37.9 %	73.6 %
Caries	0.30	25.8 %	84.1 %
Periapical lesion	0.15	39.9 %	46.8 %
impacted tooth	0.15	72.0 %	77.1 %

Table 10: Tooth-level evaluation of the identical predictions. The caries detector is correct in 74 % of the cases where it fires.

Protocol. Before comparing detectors across corpora, compare the *annotations*: report the median box area per class in each, and treat any class whose ratio departs materially from unity as incomparable under IoU until the granularity is reconciled. A class on which the two corpora agree should be carried as a control, because without one there is no way to distinguish a units mismatch from a domain gap—both present as a collapse. Here the control also quantifies the real gap: impacted tooth falls from 0.490 to 0.290, a 41 % relative drop, which is the discount any cross-dataset claim must carry.

6 The Label Ceiling, Measured

The apparatus above certifies comparisons between models. This section turns the same consensus logic onto the corpus. Fifteen prediction files were on disk, spanning four architectures, seven loss configurations and three seeds. For every ground-truth instance of the three floor classes we ask the most lenient question available: did *any* model place a box of the right class at even 0.10 IoU, at the frozen 0.15 operating point?

class	missed by all 15	total GT	share
Periapical lesion	343	799	**42.9 %**
Bone Loss	156	473	**33.0 %**
Caries	412	1615	**25.5 %**

Table 11: 911 annotations that fail the most lenient detection criterion under every model trained on this corpus. These are the classes at the metric floor.

An instance that fails this is not a hard example; it is an absent one. Two follow-ups pin down what the set is. First, holding the resolution model out of the consensus and asking how much of the set it recovers: the model that improved caries by 35 % recovers **2.7 %** of the universally missed instances. They are therefore not small findings starved by downsampling—the one intervention proven to help small findings barely touches them. Second, the external check cuts the other way on the labels as a whole: zero-shot on a professionally annotated corpus the same detector is 73.6–84.1 % precise at tooth level, so

model	mAP	AP50	AP75	tail
baseline	0.1051	0.2590	0.0687	0.0365
boundary alone	0.1071	0.2585	0.0680	0.0441
weighting + boundary	0.1007	0.2430	0.0650	0.0261
weighting alone	0.0973	0.2369	0.0612	0.0252

Table 12: Segmentation, held-out test. The nominal $+0.21$ pp of the boundary arm decomposes to three classes with 4, 8 and 8 test instances.

arm	mAP	tail
baseline	0.1625	0.0368
denoising only	0.1633	0.0455
plain reweighting	0.1461	0.0270
unified ($\tau = 1.0$)	**0.1020**	**0.0000**

Table 13: Detection, validation, matched budget. The unified treatment loses 4.41 pp to plain reweighting, which loses 1.64 pp to changing nothing. Tail AP is exactly zero in all unified arms.

the label set is not uniformly unreliable. The 911 are a specific, enumerable subset, and the list itself is the audit's sampling frame.

Protocol. Consensus failure across independently trained models is a label-error detector that costs nothing beyond predictions already held. Run it over the *training* split with models trained on that split and the signal sharpens further: an annotation a model cannot fit after being optimised to fit it is the strongest cheap evidence of label error available. No architecture decision on such a corpus is safe until this bound is known, because below it, apparent model failures are label failures.

7 Results

The τ sweep gives -6.05, -2.85, -0.24 pp at $\tau = 1.0, 0.5, 0.25$: monotonic recovery, never reaching baseline. The catastrophe is a mis-scaled constant; the deficit is not.

Divergence is patterned. Every arm applying the adjustment inconsistently diverged with `inf` or `NaN` cost matrices in the Hungarian matcher; every consistent arm was stable. Consistency prevents divergence without producing accuracy.

8 The Mask Deficit Is Coefficient Prediction, Not Representation

The 20-point gap between what the model achieves (Dice 0.6969) and what its prototype grid allows (0.8963) had no attribution. Prototype-based heads reconstruct each mask as $\sigma(\mathbf{c}^\top P)$, so either the learned 32-dimensional basis cannot represent these shapes, or the basis is adequate and the coefficient head fails to locate the right point in it. Those call for opposite fixes.

We separate them without training. For every ground-truth instance we solve, in closed form, for the coefficient vector that best reconstructs it from the model's own prototypes,

$$\mathbf{c}^* = \arg\min_{\mathbf{c}} \|P^\top \mathbf{c} - y\|^2, \tag{1}$$

over the instance's box crop, where $y \in \{-1, +1\}$ is the ground-truth mask. Thresholding $P^\top \mathbf{c}^*$ gives the best mask *any* coefficient head could produce from that basis.

quantity	Dice
grid ceiling (what the resolution allows)	0.8963
oracle coefficients, full resolution	**0.8659**
oracle coefficients, at prototype resolution	0.9636
what the trained model achieves	0.6969

source of the 0.1994 gap	Dice lost	share
prototype **basis** cannot represent it	0.0304	**15 %**
coefficient head fails to find it	0.1690	**85 %**

Table 14: 4123 instances. The basis is not the constraint: handed optimal coefficients the model jumps from 0.70 to 0.87.

This closes the loop on every negative result above. Boundary terms and comparator losses reshape the *objective*; higher prototype resolution enlarges the *representation*. Neither is the binding constraint, which is why neither moved the number, and why the P2-fed head performed worse rather than better—it added resolution to a basis that was never the limit.

The head capacities make this concrete. In YOLOv8x-seg the coefficient branch bottlenecks 320 to 80 channels and carries 1.33 M parameters, against 2.26 M for the prototype generator and 7.41 M for classification. The smallest of the three heads is responsible for 85 % of the mask deficit.

8.1 The Attribution Is Correct and Not Actionable

A localisation earns its keep by telling you what to change, so we tested both things it implies. Capacity: the coefficient branch rebuilt from 80 to 256 hidden channels. Supervision: an auxiliary term pulling the predicted coefficients toward $\mathbf{c}^*$ itself, recomputed each step from the model's own prototypes and detached.

Neither improves masks. The capacity arm is the $+0.64$ pp result of Section 5.3, which the box control shows to be detection-side; on the paired common set its masks are flat, and the only separable contour metric favours the reference. The supervision arm is -0.70 pp with paired Dice -0.0017, and its own pixel BCE at epoch 50 is roughly 3.05 against the reference's 1.65: the auxiliary term pulled the head away from the pixel optimum rather than toward it. A scale mismatch would be the natural explanation and is not the right one—target and prediction agree to within 3 % in RMS. The coefficients minimising pixel BCE and those best reconstructing the mask in least squares are different points.

A calibration failure worth recording. The first supervision run was numerically broken and still produced a publishable-looking number. The normal-equation matrix $A = PBP^\top$ has eigenvalues spanning 10^{-6} to 10^3 on real instances, so a fixed ridge regularises some and leaves others singular; training loss reached 7.2×10^7 with NaN at epoch 10, and the arm still reported mAP 0.1106. The weight had been calibrated on the *converged* model, where $\overline{\mathbf{c}^{*2}} = 2.86$, then applied to a run starting from COCO weights where it measures 4 to 53. Calibrating a hyper-parameter on a model state the run never occupies is its own failure mode, and a loss curve is the cheapest detector of it: the metric alone looked unremarkable.

9 Noise Floor (Measured)

The reference configuration was trained at three seeds, everything else identical. Determinism is exact on this pipeline, so seed choice is provably the only source of variation.

seed	mAP	AP50	AP75	head	tail
42	0.1055	0.2549	0.0661	0.2815	0.0101
1337	0.1056	0.2573	0.0679	0.2837	0.0117
2024	0.1037	0.2568	0.0629	0.2823	0.0105
sd	**0.0010**	0.0013	**0.0025**	0.0011	0.0008

Table 15: Noise floor at 2 sd: mAP ± 0.21, AP50 ± 0.26, AP75 ± 0.50, head ± 0.22, tail ± 0.17 pp.

Three consequences, and they are not the same verdict. The seven-arm ablation spans a 0.36 pp range, *inside* the band—selecting a winner on mAP was selecting noise. The best segmentation arm on test is $+0.21$ pp, sitting *exactly at* the threshold. But the isolated boundary effect on AP75 is $+0.70$ pp against a ± 0.50 pp floor, so it **exceeds** noise on validation at 50 epochs, and then reverses on test at 100 epochs. That is a **transfer failure, not a noise result**: the objective does what its mechanism predicts under the conditions it was measured in, and that does not survive a change of split and schedule.

10 Discussion

The standard long-tail and boundary-aware toolkit does not transfer to this corpus, and most of the reasons are not about the methods. Two are about the data: it cannot be evaluated as shipped, and 16 of 31 classes lack the support to measure anything. Two are about measurement: distance metrics and group means both produce confident wrong answers under conditions easy to satisfy accidentally. One is about regime: a published constant validated at 100:1 does not survive 34,320:1.

Only one is about a method—the band-Dice objective produces the AP75/AP50 signature its mechanism predicts, and that signal does not survive to test.

We think the protocols of Sections 3.1, 5.1 and 5.2 are the durable output. Each is cheap, each applies to any long-tailed clinical segmentation study, and each caught an error in our own work that would otherwise have been published.

11 Limitations

Single seed on all reported arms; the three-seed noise floor is measured above. Single corpus, single backbone family per task. No comparison against prior published results, because none exist for this dataset. No external benchmark, and therefore no long-tail claim in the title, abstract or conclusions. Two detection cells are recorded as diverged rather than scored.

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